CURRICULUM VITAE

Alex Wickes, RA

RESEARCH ARCHITECT, LEED AP BD+C

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From a journey across leading and designing master plan developments as a licensed architect, to exploring the introduction of additive manufacturing into professional practice, one thing is in common across the industry: curiosity of the future, and a desire to live it. The process of researching, developing, and showcasing design methods involves stakeholder involvement and targeted outreach, to develop long term partnerships with aligned values and goals.

Driving Thesis:

In this world, every process involves giving and taking. Each method of creation has its own advantages and disadvantages; destruction is necessary for creation, and there are inherent limitations to the act of creating. In the modern world, the most significant limitation of creation is time, our most valuable resource. While it may be possible to purchase time on a superficial level, it remains predominantly unable to be bought. When discussing creation and its temporal aspect, another crucial factor to consider is complexity. In traditional making, complexity and cost have a 'ying and yang' relationship, to have one, you must accept the other.

Unlike traditional making methods, Additive Manufacturing is unique in its formlessness. This means that the complexity of creation is not tied to cost. This reverts the fundamental design vs construction challenge where building things complex is building things expensive.

This new dynamic between complexity and cost brings forth the essential elements of my personal thesis. In this approach, additive manufacturing, augmented reality and artificial intelligence are not means to an end, but a combination that can bring our built environment to its fullest potential through a Synthographic Process. This is my search.

ARTIFICIAL INTELLIGENCE:
Gap: There are solutions to complexities that surpass our human capacity of understanding and creativity.

AUGMENTED REALITY:
Gap: We are unable to 'see' and interact these solutions and require a new medium to comprehend them.

ADDITIVE MANUFACTURING
Gap: While we can now understand and see these solutions, they are unable to be built by traditional making methods.

THE HYPER TILE 19-02 (12'-0" x 75'-0")
STABLE DIFFUSION | PYTHON | PROCREATE

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Philosophical Lens

Approaching architecture to create storytelling experiences backed by data and research.

Technical Aptitude Adobe Suite | MS Office | Python | C# | Swift |

RHINOCEROS 3D/GRASSHOPPER
Karamba3D/Fenix | Lattice & Additive Manufacturing | LadyBugTools | Generative Design

ARTIFICIAL INTELLIGENCE
Stable Diffusion | MidJourney | Lunchbox ML | ChatGPT | TensorFlow

AUGMENTED/VIRTUAL REALITY
Unity3D | Unreal Engine | iOS AR Kit/USDZ | Android ARCore | Microsoft HololLens | Apple VisionPro

ENVIRONMENT DEVELOPMENT
3D Printing, CNC, Laser Cutter, shop tools | LIDAR, Photogrammetry | Blender | Houdini

AUTODESK
Revit | RhinoInside | Dynamo | 3DSMax | Fusion360 | Maya

Selected Experience: 8+ years 15 years in AEC, RA, Lead AP BD+C

PFLUGER - ARCHITECTURAL R&D APPLICATION LEAD RESEARCH ARCHITECT 2024-PRES
Research & Development of Artificial Intelligence, and immersive experiences. Development and integration of Business development and operational software. Sectors: K-12 and Higher Education.

CORGAN - ADDITIVE MANUFACTURING & MASS TIMBER HUGO: DESIGN RESEARCHER 2023-2024
Future of The AEC Executive Leadership Team Advisement. Additive Manufacturing & Mass Timber Practice Integration, Curiosity Report. Sectors: Aviation, Data Centers, Education, Media Lab, Shop.

GFF - HENDERSON MIXED-USE PROJECT ARCHITECT & DESIGNER 2021-2023
Artificial Intelligence, Sustainability, Computational Design, and Advanced Construction in the AEC. Sectors: Lifestyle & Restaurant, Office, Planning, Multi-Family. Committees

JMA - EVERGY ELECTRIC EMPORIA HEADQUARTERS PROJECT DESIGNER 2020-2021
Practice Sectors: Office and Industrial: Project management, Project designer, LEED Projects

K&C - ENMARKET PROTOTYPE DESIGN PROJECT DESIGNER & REVIT INSTRUCTOR 2016-2018
Practice Sectors: Office, Lifestyle & Restaurant, Historic Preservation, BIM Management

H&H - CONSTRUCTION AND LAND SURVEYING SURVEYOR 2013-2016
CITY OF AUGUSTA - HOUSING AND URBAN DEVELOPMENT PROJECT DESIGNER 2012-2013

Background: Master of Architecture

WASHINGTON UNIVERSITY IN SAINT LOUIS SAINT LOUIS MO 2018-2020
Master of Architecture with Honors | Research Assistant | Co-PI

SAVANNAH COLLEGE OF ART AND DESIGN SAVANNAH GA, 2013-2017
Bachelor of Architecture Cum Laude | Minor in Architectural History

DALLAS ZOO DISTINGUISHED VOLUNTEER 2022-2024
Environmental Research & Conservation | Special Events | Zoo Lights | Member Mornings

PUBLICATIONS

Wickes, A., Desooky, A., Flores, S., Hoelting, M. (2024) Mass Timber Design Integration: Data Centers. Corgan.

Wickes, A., (2024). Additive Manufacturing Design. In S. Flores, M. Hoelting, *Executive Leadership Team Advisement: The 5 Year Future of The AEC*. Corgan.

Wickes, A., Desooky, A. (2024). Mass Timber Design. In S. Flores, M. Hoelting, *Executive Leadership Team Advisement: The 5 Year Future of The AEC.* Corgan.

Wickes, A., (2024). Material and Construction Trends. In S. Flores, M. Hoelting, *CorganOne: The 5 Year Future of The AEC*. Corgan.

Wickes, A., Flores, S., Hoelting, M. (2024). Additive Manufacturing: Practice Integration. In *CorganOne: Hugo Research.* Corgan.

Wickes, A., Hoelting, M. (2024). Research Literature Database. In *CorganOne*. Corgan.

Wickes, A., Hoelting, M., Flores, S. (2023) Curiosity Report: Additive Manufacturing. In *2023 Curiosity Report*. Corgan.

Han, D., Yin, H., Qu, M., Zhu, J., Wickes, A. (2019). Technical Analysis and Comparisons of Formwork-making Methods for Customized Prefabricated Buildings: 3D Printing and Conventional Methods. In *Journal of Architectural Engineering*. doi: 0.1061/(ASCE) AE.1943-5568.0000397.

Wickes, A. (2022). Unreal Engine and City Engine. In C. Wacta. *Resilient Design: Rethinking Architecture with GIS*. Environmental Systems Research Institute, Inc Publishing, 2023.

Wickes, A., Yin, H., Qu, M. (2019) Co-PI: Autonomous Onsite 3D Printing and Robotic Building Construction (A-3D-PRo). In, *Department of Energy: Advanced Building Construction*. FOA-02099.

Wickes, A., Yin, H., Qu, M. (2019). Co-PI: Adversarial- machine-learning aware adaptive building (AMLAB) controls for cyber-security and optimal operation. In, *Department of Energy: Buildings Energy Efficiency Frontiers & Innovation Technologies (BENEFIT).* FOA-0002090.

Wickes, A. (2022). Portraits. Business Council For The Arts, https://ntbca.org/omot.

Wickes, A. (2022). Portraits: A four series of acrylic works. In *On My Own Time Regional Winner's Exhibit at NorthPark Center Dallas Texas*.

Wickes, A. (2021). GFF - The Iterative Simulation Design Process.

Wickes, A. (2020). Selected works Portfolio. Issuu Digital Publishing, https://issuu.com/alexwickes/docs/wickes_ portfolio_2020.

Wickes, A. (2019). 3D Printing Parts and Types. To be published.

Wickes, A. (2019) ADPRU. In, *2018-2019 Graduate School of Architecture & Urban Design at Washington University in St. Louis,* pp. 144-147. Washington University In Saint Louis.

Wickes, A, Ahmadi, A., Bock, L., Poole, G. (2017) The Future of GIS with VR/AR/MR and 3D printing. In, *ESRI 2017 Federal GIS Conference, Walter E. Washington Convention Center*, Washington, DC.

Wickes, A. (2019) CITY CLUE. In *2018-2019 Graduate School of Architecture & Urban Design at Washington University in St. Louis*, pp. 344-347. Washington University In Saint Louis.

Wickes, A. (2017) SUBLEVO: Net Zero Research Laboratory. In *2016-2017 SCAD Academic Catalogue*, pp. 78-79. Savannah College of Art and Design.

Wickes, A. (2015). The Wharf. Autodesk Featured Rendering, https://gallery.autodesk.com/a360rendering/projects/17031/the- wharf. Autodesk.

Wickes, A. (2014). The Wharf. In A*utodesk Panoramic Rendering.* https://pano.autodesk.com. Autodesk.

Wickes, A. (2014). Subway. In *Autodesk Panoramic Renderings.* https://pano.autodesk.com. Autodesk.

HONORS

WashU alum events

prism AIA Emerging Leaders Program Guest Panel Speaker: The Age of AI in Architecture.

Mass enginnereed timber pres to firm

C suite Special Advisor on AI assisted architecture buiess practices.

Mod 2 publication with Sunwest

Sanctuary sapces publication with snuwest

Tasa tasb - showcased 12'x6' physical model at 2025 tasa tasb. riftsawn white oak and pla/resin. 60' 12v Cobod led in aluminum diffuser Chanels.

urban land instutue and Coaa memeber and volenteer

State of firm presentation on immersive learning, Pfluger Prism, RB Reboot

Tasa presentation AIA Emerging Leaders Program Guest Panel Speaker: The Age of AI in Architecture.

CPISD Immersive learning Presentation 2024 AIA Emerging Leaders Program Guest Panel Speaker: The Age of AI in Architecture.

2024 AIA Emerging Leaders Program Guest Panel Speaker: The Age of AI in Architecture.

2024 AIA Emerging Leaders Program Guest Panel Speaker: The Age of AI in Architecture.

2023 Introduction of iOS Augmented Reality Additional Service stream to GFF.

2022 AIA Emerging Leaders Program: The Dreyfus Club at White Rock Lake and class graduate.

2022 "12" x 12" Portraits" exhibited at the 2022 On My Own Time Regional Winner's Exhibit at NorthPark Center. First Place - Amateur Artist Works on Canvas.

2022 Young Guns member and TREC Volunteer for Fight Night.

2021 GFF University presentation on the Iterative Simulation Design Process in RhinoGrasshopper.

2020 LEED Advanced Professional: Building Design and Construction.

2020 LEED Green Associate.

2020 Master of Architecture with Honors - Washington University in Saint Louis.

2020 William Tao Prize - Washington University in Saint Louis Sam Fox School of Design All School Award: excellence in the understanding and application of building systems in architecture, including illumination, electrical and mechanical engineering, and energy-efficient design.

2019 Department of Energy (DOE): Adversarial-machine-learning aware adaptive building (AMLAB) controls for cyber security and optimal operation FOA-0002090.
2019 Department of Energy (DOE) Co-PI: Autonomous Onsite 3D Printing and Robotic Building Construction (A-3D-PRo) FOA-02099.

2019 Co Author of publication in JAE: "Technical Analysis and Comparisons of Formwork-making Methods for Customized Prefabricated Buildings: 3D Printing and Conventional Methods." doi: 0.1061/(ASCE)AE.1943-5568.0000397.

2019 Work featured and displayed in The Saint Louis Kemper Art Museum during the Sam Fox School of Design and Visual Arts Fall Scholarship Reception.

2019 Works featured in Washington University in Saint Louis Fall Symposium: Decoys and Depictions.

2019 Works featured in Washington University Approach 2018-2019.

2018 Recipient of Directors Scholarship at Washington University.

2018 Solar Decathlon Building Science Certificate, RDH Building Science Laboratories.

2017 Bachelor of Fine Arts, Architecture & Minor in Architectural History Savannah College Of Art and Design.

2017 "VR/AR and 3D printing of GIS Data", Exhibitor and Student Ambassador, ESRI 2017 Federal GIS conference in Washington DC.

2017 Autodesk Building Performance Analysis Certificate Recipient.

2016 International Design Award, "Savannah Rowlocks, Savannah GA", Architecture-New Commercial Building: Bronze. 2016 Works featured as banner on A360 Gallery Homepage.

2016-20 Works selected as Featured Renderings for Autodesk Cloud Rendering Services
2016-20 Works selected for marketing of Autodesk Panoramic Rendering
2014-20 Autodesk Student Expert
2013-17 Academic Honors - Savannah College of Art and Design

2013 Associates of Liberal Arts from Armstrong Atlantic State University 2009-13 Academic Honors - Armstrong Atlantic State University.

2009-13 Academic Honors - Armstrong Atlantic State University.

Application and Software Development:

2025 Pfluger Prism - Cost estimation web applicaiton powered by Vermulens. SQL php back end, React front end web app, Pfluger Vision pipeline

2025 Pfluger Vision - Business development dashboard, Vantage point integration, SQL php back end, React front end web app, Pfluger Prism pipeline

2025 Pfluger Lens - Immersive platform with Elumenati Omniwarp

2024 Seed Stone - Chritian faith based mindlufless and meditation iOS application

2024 Additive Manufacturing Model Conversion

2023 iOS GFF Augmented Reality Client Presentation Apps

2023 Founding of Blue Mirage: Augmented Reality GIS solutions for urban design.

2020 RevTAR Release: MCU (Manifestation Creation Unit) with marker-less AR tabletop, site-scale walk-through and component inspector

2019 RevTAR Release: Real-Time Custom 3d Printing machine controls for a segmented telescopic arm

2019 RevTAR Release: Real-Time Robotic Arm 3D Printing Control/Monitoring/Slicing/Building Design Tool

2019 RevTAR Release: Real-Time 3D Printed Full Scale Building Design Tool with marker-less Augmented Reality

2018 RevTAR Release: Real-Time 4 dimensional architectural voxel sculpting tool"

2017 RevTAR Release: Real-Time BIM CD Coordination and Review Tool with marker based Augmented Reality for Autodesk Revit

2017 Founding of RevTAR: Augmented reality solutions for architectural design.

"The Squeezed City"
36" x 60" Acrylic on Canvas

1:1 fabrication mock up of a 236' canopy with commissioned Synthographic artwork
Translucent polycarbonate, Stable Diffusion, Python, Procreate, USDZ

Mass Timber Data Centers
Revit, Grasshopper, Finex3D